Redundant Campus Network

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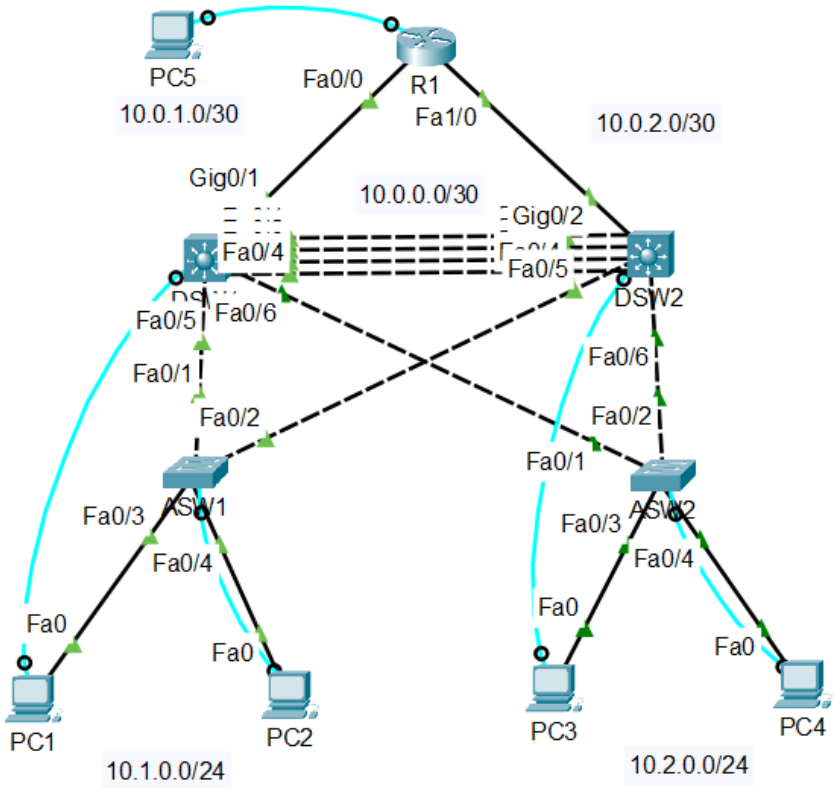
Overview

For this project I designed and built a small redundant campus network in Cisco Packet Tracer. My goal was to practice a proper three-tier design, access switches at the bottom, multilayer distribution switches in the middle, and a router on top, and to make the network survive a single uplink failure without anyone having to touch it.

The finished network keeps working when one of the distribution uplinks is cut. I verified this myself with traceroutes during a simulated link failure, which was honestly the most satisfying part of the whole project.

Topology

The design has two access switches (ASW1 and ASW2), two multilayer distribution switches (DSW1 and DSW2), and a router (R1) on top. PC1 and PC2 live in VLAN 10 behind ASW1, PC3 and PC4 live in VLAN 20 behind ASW2, and PC5 is just a console machine I use to manage R1 out-of-band, like a console cable in the real world.



· ASW1 connects PC1 and PC2 in VLAN 10 (10.1.0.0/24) and trunks up to both DSWs.

· ASW2 connects PC3 and PC4 in VLAN 20 (10.2.0.0/24) the same way.

· DSW1 and DSW2 are multilayer switches joined by a routed EtherChannel (Port-channel 1, 10.0.0.0/30).

· Each DSW has a routed /30 uplink to R1: 10.0.1.0/30 for DSW1, 10.0.2.0/30 for DSW2.

· PC5 is only there to console into R1, my out-of-band management machine.

How the design works

Each PC's default gateway is an HSRP virtual IP (.254), not a specific box, so the gateway survives a distribution switch dying. I split the work: DSW1 is the active router for VLAN 10 and DSW2 is active for VLAN 20, so both boxes carry traffic instead of one sitting idle.

Each DSW has a primary default route through its own R1 link, plus a floating static (administrative distance 10) pointing across the EtherChannel at the other DSW. R1 does the symmetric thing for return traffic: a primary route per LAN subnet plus a backup route at AD 5 through the other DSW. No single uplink failure can isolate a VLAN.

Issues I ran into

VLAN 10 could not reach R1 after cutting DSW1's uplink

This was the big one. DSW1's only way upstream was its default route via 10.0.1.2. When I cut G0/1, that route disappeared and DSW1 had no idea where to send VLAN 10 traffic, but it was still the active HSRP gateway, so it kept the gateway IP and silently dropped everything.

The fix was a floating static default route across the EtherChannel: ip route 0.0.0.0 0.0.0.0 10.0.0.2 10, with the mirror image on DSW2. The lesson stuck with me: HSRP protects the gateway, not the uplink. A switch can stay the active gateway and still blackhole traffic if its own path upstream is gone.

EtherChannel members left shut down

I configured DSW2's channel-group with the member ports shut as a precaution, but only brought up po1. The physical ports stayed shut, so the whole inter-DSW link was dead. Lesson learned: no shut the members after bundling, and always verify with show etherchannel summary instead of trusting what I typed.

Duplicate IPs on the HSRP SVIs

I originally gave both DSWs the same .1 addresses on their SVIs. In an HSRP pair only the virtual IP is shared, the real IPs must be unique. Packet Tracer actually flagged this with a %IP-4-DUPADDR warning on DSW1, and I moved DSW2 to .2 on both VLANs.

Final state

After fixing the issues above, everything is clean: DSW2's SVIs are now .2 on both VLANs so there are no duplicate-address warnings, the EtherChannel members are up on both sides, each DSW has a floating static default route through the other, and R1 has a loopback (1.1.1.1/32) acting as a pretend remote network for testing. The show standby output confirms exactly the split I wanted: DSW1 active for VLAN 10, DSW2 active for VLAN 20.

Results

- Inter-VLAN routing works: VLAN 10 and VLAN 20 hosts reach each other.
- All PCs reach R1 and its loopback (1.1.1.1).
- With DSW1's uplink cut, VLAN 10 traffic fails over through DSW2 with no manual intervention. The traceroute visibly shows the path going through 10.1.0.1, then 10.1.0.2, then reaching 1.1.1.1.

· Hosts get DHCP leases from .11 upward with .254 as their gateway.

What I learned

· A redundant topology does not fail over by itself. Every device still needs a route telling it where to send traffic when its preferred path dies.

· HSRP protects the gateway (.254), not the uplink. A switch can stay the active gateway and still blackhole traffic if its own path upstream dies.

· Floating statics (AD > 1) are a clean way to do failover without a routing protocol.

· Small slips like a forgotten no shut or a wrong IP silently break redundancy. Always verify with show commands instead of trusting the config I typed.

Appendix: Final Configurations

ASW1

```
en
conf t
ho ASW1
no ip domain-lookup
int range f0/3-4
switchport mode access
switchport access vlan 10
ex
int range f0/1-2
switchport mode trunk
switchport trunk allowed vlan 10
switchport trunk native vlan 99
ex
ip dhcp snooping
ip dhcp snooping vlan 10
int f0/1
ip dhcp snooping trust
no ip dhcp snooping information option
ex
do copy run start
```

ASW2

```
en
conf t
ho ASW2
no ip domain-lookup
int range f0/3-4
switchport mode access
switchport access vlan 20
int range f0/1-2
switchport mode trunk
switchport trunk allowed vlan 20
switchport trunk native vlan 99
ex
ip dhcp snooping
ip dhcp snooping vlan 20
int f0/1
ip dhcp snooping trust
no ip dhcp snooping information option
```

```
ex
do copy run start
```

DSW1

```
en
conf t
ho DSW1
no ip domain-lookup
ip routing
int range f0/1-4
no sw
channel-group 1 mode on
int po1
ip add 10.0.0.1 255.255.255.252
no shut
int f0/5
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10
switchport trunk native vlan 99
int f0/6
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 20
switchport trunk native vlan 99
ex
vlan 10
vlan 20
int vlan 10
ip add 10.1.0.1 255.255.255.0
no shut
standby ver 2
standby 10 ip 10.1.0.254
standby 10 priority 150
standby 10 preempt
ex
int vlan 20
ip add 10.2.0.1 255.255.255.0
no shut
standby ver 2
standby 20 ip 10.2.0.254
standby 20 priority 50
standby 20 preempt
ex
ip dhcp exclude 10.1.0.1 10.1.0.10
ip dhcp exclude 10.2.0.1 10.2.0.10
ip dhcp pool POOL1
network 10.1.0.0 255.255.255.0
default-router 10.1.0.254
domain-name sean.com
ex
ip dhcp pool POOL2
network 10.2.0.0 255.255.255.0
default-router 10.2.0.254
domain-name sean.com
ex
```

```
int g0/1
no sw
ip add 10.0.1.1 255.255.255.252
no shut
ex
ip route 0.0.0.0 0.0.0.0 10.0.1.2
ip route 0.0.0.0 0.0.0.0 10.0.0.2 10
do copy run start
```

DSW2

```
en
conf t
ho DSW2
no ip domain-lookup
ip routing
int range f0/1-4
no sw
shut
channel-group 1 mode on
int po1
ip add 10.0.0.2 255.255.255.252
no shut
int range f0/1-4
no shut
int f0/5
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10
switchport trunk native vlan 99
int f0/6
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 20
switchport trunk native vlan 99
ex
vlan 10
vlan 20
int vlan 10
ip add 10.1.0.2 255.255.255.0
no shut
standby ver 2
standby 10 ip 10.1.0.254
standby 10 priority 50
standby 10 preempt
int vlan 20
ip add 10.2.0.2 255.255.255.0
no shut
standby ver 2
standby 20 ip 10.2.0.254
standby 20 priority 150
standby 20 preempt
ex
int g0/2
no sw
ip add 10.0.2.1 255.255.255.252
no shut
```

```
ex
ip route 0.0.0.0 0.0.0.0 10.0.2.2
ip route 0.0.0.0 0.0.0.0 10.0.0.1 10
do copy run start
```

R1

```
en
conf t
ho R1
no ip domain-lookup
int l0
ip add 1.1.1.1 255.255.255.255
no shut
int f0/0
ip add 10.0.1.2 255.255.255.252
no shut
int f1/0
ip add 10.0.2.2 255.255.255.252
no shut
ex
ip route 10.1.0.0 255.255.255.0 10.0.1.1
ip route 10.1.0.0 255.255.255.0 10.0.2.1 5
ip route 10.2.0.0 255.255.255.0 10.0.2.1
ip route 10.2.0.0 255.255.255.0 10.0.1.1 5
do copy run start
```